AW862XX 软件设计指南

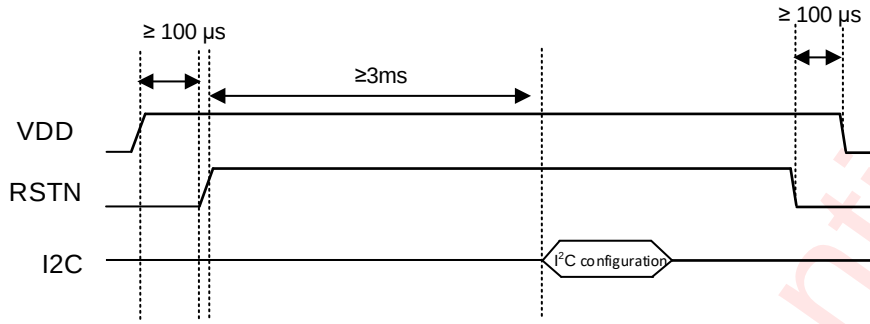
1 AW862XX 简介

AW862XX 是一款常压线性马达驱动芯片、具有基于反向电动势的 F0 检测和跟踪功能、集成了 3KB SRAM 用于用户定义波形，支持 3 个波形采样率(12K/24K/48K)。本文档旨在帮助客户快速构建 AW862XX 的驱动程序，能够支持简单的应用场景，占用较少资源。

2 软件设计

2.1 硬件复位

1. RSTN 拉低 2ms 后再拉高 8ms，完成硬件复位，参考时序图如下：



2. 参考代码：

```
HAL_GPIO_WritePin(GPIOB, GPIO_PIN_8, 0);
HAL_Delay(2);
HAL_GPIO_WritePin(GPIOB, GPIO_PIN_8, 1);
HAL_Delay(8);
```

2.2 Chipid 校验

1. AW862XX系列芯片的chipid都存放在寄存器0x00和0x64中，通过读取寄存器可以获取当前IC的chipid（AW86224与AW86225不做区分）：

IC	ADDR	Default	ADDR	Bit6	Bit0
AW86214	0x00	0x01	0x64	1	1
AW86223	0x00	0x00	0x64	0	1
AW86224	0x00	0x00	0x64	0	0
AW86225	0x00	0x00	0x64	0	0

2. 参考代码：

```
uint8_t reg[2] = {0};


haptic_nv_i2c_read(0x00, &reg[0]);
switch (reg[0]) {
case 0x00:
    haptic_nv_i2c_read(0x64, &reg[1]);
    if ((reg[1] & 0x41) == 0x01) {
        printf("aw86223 detected");
    }
    if ((reg[1] & 0x41) == 0x00) {
        printf("aw86224 or aw86225 detected");
    }
case 0x01:
    haptic_nv_i2c_read(0x64, &reg[1]);
    if ((reg[1] & 0x41) == 0x41) {
        printf("aw86214 detected");
    }
...
}
```


2.3 软件复位

1. 在识别到chipid 为四款IC 的其中之一后，可以对IC 进行软件复位操作，向0x00 寄存器写0xAA 即

可；

2. 参考代码：

```
haptic_nv_i2c_write(0x00, 0xAA);
```

2.4 芯片功能初始化

2.4.1 波形采样率

1. 通常使用 12K 波形采样率;

2. 参考代码

```
haptic_nv_i2c_write_bits(0x44, ~(3<<0), (2<<0));
```

2.4.2 保护模式

1. 直流保护功能通过寄存器 0x4E 和 0x4F 配置，PRLVL 和 PRTIME 可以默认保持，初始化为启用;

2. 参考代码

```
haptic_nv_i2c_write_bits(0x4E, ~(1<<7), (1<<7));
```

2.4.3 Gain_Bypass

1. Gain bypass 功能允许芯片在振动期间修改增益值，初始化为启用;

2. 参考代码：

```
haptic_nv_i2c_write_bits(0x49, ~(1<<6), (1<<6));
```

2.4.4 其他

```
haptic_nv_i2c_write_bits(0x2D, ~(1<<4), (1<<4));//使能 3K SRAM
haptic_nv_i2c_write_bits(0x2D, ~(1<<5), (1<<5));
haptic_nv_i2c_write_bits(0x3A, ~(3<<3), (2<<3));
haptic_nv_i2c_write_bits(0x77, ~(3<<6), (3<<6));
```

2.5 RAM 初始化

AW862XX 系列芯片拥有3KB RAM，FIFO 最大可达2KB，使用RAM 模式或RTP 模式前，需要对芯片的RAM 进行初始化配置。

2.5.1 数字模块时钟

1. 初始化 AW862XX 芯片的 RAM 之前，需要先打开数字模块的时钟，初始化完成再关闭即可；

2. 参考代码：

```
haptic_nv_i2c_write_bits(0x43, ~(1<<3), (1<<3)); //Raminit enable
HAL_Delay(1);
……                                                    //RAM config
haptic_nv_i2c_write_bits(0x43, ~(1<<3), (0<<3)); //Raminit disable
```

2.5.2 RAM基地址

1. 通过设置RAM 的基地址来划分FIFO 和RAM 空间的大小。默认使用2KB FIFO，剩余1KB 为RAM 波形存储空间;

2. 参考代码:

```
haptic_nv_i2c_write_bits(0x2D, ~(0x0F<<0), 0x08);
haptic_nv_i2c_write(0x2E, 0x00); // The RAM base addr starts from 0x0800, the FIFO size is 2KB
haptic_nv_i2c_write(0x40, 0x08);
haptic_nv_i2c_write(0x41, 0x00); // RAM pointer set to start from scratch
```

2.5.3 RAM数据初始化

1. 初始化RAM 基址后，可以将RAM 波形数据写入并存储在芯片RAM 中，并且在上电时只需写入一次RAM 波形数据；

2. 参考波形数据:

```
uint8_t aw862xx_ram_data[] = { //Default 170HZ waveform
0x55,0x08,0x11,0x09,0x76,0x09,0x77,0x0a,0x80,0x0a,0x81,0x0b,0x2e,0x0b,0x2f,0x0b,
0x61,0x00,0x0e,0x1d,0x2b,0x38,0x45,0x50,0x5b,0x63,0x6b,0x71,0x75,0x77,0x77,0x76,
0x73,0x6e,0x68,0x5f,0x56,0x4b,0x3f,0x32,0x24,0x16,0x07,0xfa,0xeb,0xdd,0xcf,0xc2,
0xb6,0xab,0xa1,0x99,0x92,0x8d,0x8a,0x89,0x89,0x8b,0x8f,0x95,0x9c,0xa5,0xaf,0xba,
0xc7,0xd4,0xe2,0xf1,0x00,0x0d,0x1c,0x2a,0x37,0x44,0x50,0x5a,0x63,0x6a,0x70,0x74,
0x77,0x77,0x76,0x73,0x6e,0x68,0x60,0x57,0x4c,0x40,0x33,0x25,0x17,0x08,0xfb,0xec,
0xde,0xd0,0xc3,0xb6,0xab,0xa2,0x99,0x93,0x8e,0x8a,0x89,0x89,0x8b,0x8f,0x94,0x9c,
0xa4,0xae,0xba,0xc6,0xd3,0xe1,0xf0,0xff,0x0c,0x1b,0x29,0x37,0x43,0x4f,0x59,0x62,
0x6a,0x70,0x74,0x77,0x77,0x76,0x73,0x6f,0x69,0x61,0x57,0x4c,0x41,0x34,0x26,0x18,
0x09,0xfb,0xed,0xde,0xd1,0xc3,0xb7,0xac,0xa2,0x9a,0x93,0x8e,0x8a,0x89,0x89,0x8b,
0x8e,0x94,0x9b,0xa4,0xae,0xb9,0xc5,0xd3,0xe1,0xef,0x00,0xf3,0xe5,0xd8,0xcb,0xbf,
0xb4,0xaa,0xa2,0x9a,0x95,0x91,0x8e,0x8e,0x8e,0x91,0x95,0x9b,0xa3,0xab,0xb5,0xc0,
0xcd,0xd9,0xe7,0xf4,0x01,0x0f,0x1d,0x2a,0x37,0x43,0x4e,0x57,0x60,0x67,0x6c,0x70,
0x72,0x72,0x71,0x6e,0x6a,0x64,0x5c,0x53,0x49,0x3e,0x32,0x25,0x17,0x0a,0xfd,0xef,
0xe1,0xd4,0xc7,0xbc,0xb1,0xa8,0x9f,0x99,0x93,0x90,0x8e,0x8e,0x8f,0x92,0x97,0x9d,
```

0xa5,0xae,0xb8,0xc4,0xd0,0xdd,0xea,0xf8,0x05,0x13,0x21,0x2e,0x3a,0x46,0x50,0x5a,
0x62,0x68,0x6d,0x71,0x72,0x72,0x71,0x6d,0x68,0x62,0x5a,0x51,0x46,0x3b,0x2e,0x21,
0x14,0x00,0xf9,0xf2,0xeb,0xe4,0xde,0xd8,0xd3,0xcf,0xcb,0xc8,0xc6,0xc5,0xc5,0xc5,
0xc7,0xc9,0xcc,0xd1,0xd5,0xdb,0xe1,0xe7,0xee,0xf5,0xfd,0x03,0x0a,0x11,0x18,0x1f,
0x25,0x2a,0x2f,0x33,0x37,0x39,0x3b,0x3b,0x3b,0x3a,0x38,0x35,0x32,0x2d,0x28,0x23,
0x1c,0x16,0x0f,0x07,0x00,0x00,0x00,0x00,0x00,0x00,0x00,0x00,0x00,0x00,0x00,0x00,
0x00,0x00,0x00,0x00,0x00,0x00,0x00,0x00,0x00,0x00,0x00,0x00,0x00,0x00,0x00,0x00,
0x00,0x00,0x00,0x00,0x00,0x00,0x00,0x00,0x00,0x00,0x00,0x00,0x00,0x00,0x00,0x00,
0x00,0x00,0x00,0x00,0x00,0x00,0x00,0x00,0x00,0x00,0x00,0x00,0x00,0x00,0x00,0x00,
0x00,0x00,0x07,0x16,0x24,0x32,0x3f,0x4b,0x56,0x5f,0x67,0x6e,0x73,0x76,0x77,0x77,
0x75,0x71,0x6b,0x64,0x5b,0x51,0x45,0x39,0x2b,0x1d,0x0f,0x00,0xf2,0xe4,0xd6,0xc8,
0xbc,0xb0,0xa6,0x9d,0x95,0x90,0x8b,0x89,0x89,0x8a,0x8d,0x92,0x98,0xa0,0xaa,0xb5,
0xc1,0xce,0xdb,0xea,0xf8,0x06,0x15,0x23,0x31,0x3e,0x4a,0x55,0x5f,0x67,0x6e,0x73,
0x76,0x77,0x77,0x75,0x71,0x6b,0x64,0x5b,0x51,0x46,0x39,0x2c,0x1e,0x10,0x01,0xf3,
0xe5,0xd7,0xc9,0xbc,0xb1,0xa6,0x9d,0x96,0x90,0x8c,0x89,0x89,0x8a,0x8d,0x91,0x98,
0xa0,0xa9,0xb4,0xc0,0xcd,0xda,0xe9,0xf7,0x05,0x14,0x22,0x30,0x3d,0x49,0x54,0x5e,
0x66,0x6d,0x72,0x76,0x77,0x77,0x75,0x71,0x6c,0x65,0x5c,0x52,0x47,0x3a,0x2d,0x1f,
0x11,0x00,0x0f,0x1e,0x2c,0x3a,0x47,0x53,0x5e,0x67,0x6f,0x75,0x79,0x7c,0x7c,0x7b,
0x78,0x74,0x6d,0x65,0x5b,0x50,0x44,0x37,0x29,0x1a,0x0b,0xfd,0xee,0xdf,0xd0,0xc3,
0xb6,0xaa,0xa0,0x97,0x90,0x8a,0x86,0x84,0x84,0x85,0x89,0x8e,0x95,0x9d,0xa7,0xb3,
0xbf,0xcd,0xdb,0xea,0xf9,0x07,0x16,0x25,0x33,0x41,0x4d,0x59,0x63,0x6b,0x72,0x77,
0x7b,0x7c,0x7c,0x7a,0x76,0x70,0x69,0x60,0x56,0x4a,0x3d,0x30,0x21,0x13,0x00,0xfa,
0xf4,0xee,0xe9,0xe4,0xdf,0xdb,0xd7,0xd4,0xd1,0xd0,0xcf,0xcf,0xcf,0xd0,0xd2,0xd5,
0xd9,0xdd,0xe1,0xe6,0xeb,0xf1,0xf7,0xfd,0x02,0x08,0x0e,0x14,0x1a,0x1f,0x23,0x27,
0x2b,0x2d,0x30,0x31,0x31,0x31,0x30,0x2f,0x2c,0x29,0x26,0x21,0x1d,0x17,0x12,0x0c,
0x06,0x00,0x00,0x00,0x00,0x00,0x00,0x00,0x00,0x00,0x00,0x00,0x00,0x52,0x63,0x6d,0x73,
0x78,0x7a,0x7c,0x7d,0x7d,0x7e,0x7e,0x7e,0x7e,0x7e,0x7e,0x7d,0x7c,0x7b,0x79,0x76,
0x71,0x68,0x5b,0x47,0x26,0xf2,0xc8,0xae,0x9d,0x93,0x8c,0x88,0x86,0x84,0x83,0x83,
0x82,0x82,0x82,0x82,0x83,0x83,0x84,0x85,0x88,0x8b,0x91,0x9a,0xa9,0xc1,0xe6,0x07,
0x17,0x26,0x34,0x42,0x4f,0x5a,0x64,0x6d,0x74,0x79,0x7d,0x7e,0x7e,0x7c,0x78,0x72,

```
0x6a,0x61,0x56,0x4a,0x3d,0x2f,0x20,0x11,0x00,0x0d,0x1a,0x27,0x33,0x3f,0x49,0x53,
0x5b,0x62,0x67,0x6b,0x6d,0x6d,0x6c,0x6a,0x65,0x5f,0x58,0x4f,0x46,0x3b,0x2f,0x22,
0x15,0x08,0xf9,0xe9,0xda,0xcc,0xbe,0xb1,0xa6,0x9c,0x93,0x8c,0x87,0x83,0x82,0x82,
0x84,0x88,0x8e,0x96,0x9f,0xaa,0xb6,0xc3,0xd1,0xe0,0xef,0x40,0x53,0x5e,0x66,0x6b,
0x6e,0x6f,0x71,0x71,0x72,0x72,0x72,0x72,0x72,0x72,0x72,0x71,0x70,0x6f,0x6d,0x6a,
0x64,0x5c,0x4f,0x39,0x17,0x00,0x00,0x00,0x00,0x00,0x00,0x00,0x00,0x00,0x00,0x00,
0x08,0x11,0x19,0x21,0x28,0x2f,0x35,0x3a,0x3e,0x41,0x44,0x45,0x45,0x45,0x43,0x40,
0x3c,0x37,0x32,0x2c,0x25,0x1d,0x15,0x0d,0x04,0xfc,0xf4,0xec,0xe4,0xdc,0xd5,0xce,
0xc9,0xc4,0xc0,0xbd,0xbb,0xbb,0xbb,0xbc,0xbe,0xc2,0xc6,0xcb,0xd1,0xd8,0xdf,0xe7,
0xef,0xf7,
};
```

3. 参考代码：

```
haptic_nv_i2c_writes(0x42, aw862xx_ram_data, sizeof(aw862xx_ram_data));
```

2.6 工作

2.6.1 振动使能

1. RAM 短振参考代码:

```
haptic_nv_i2c_write(0x0A, 0x01); //Seq1 set RAM #1 waveform
haptic_nv_i2c_write(0x0B, 0x00); // Seq2 set stop
haptic_nv_i2c_write_bits(0x12, ~(0x0F<<4), (0<<4)); //Seq1 set loop 0，0x0F is infinite loop
haptic_nv_i2c_write_bits(0x08, ~(3<<0),(0<<0));//Enter RAM mode
haptic_nv_i2c_write_bits(0x08, ~(1<<2),(0<<2));//Close auto brake
haptic_nv_i2c_write(0x07, 0x80); //Set gain 0x80
haptic_nv_i2c_write(0x09, 0x01); //Start play
```

2. RAM 长振参考代码:

```
haptic_nv_i2c_write(0x0A, 0x04); //Seq1 set RAM #4 waveform

haptic_nv_i2c_write(0x0B, 0x00); // Seq2 set stop

haptic_nv_i2c_write_bits(0x12, ~(0xF0<<0), 0xF0); //Seq1 set loop 0x0F,infinite loop

haptic_nv_i2c_write_bits(0x08, ~(3<<0),(0<<0));//Enter RAM LOOP mode

haptic_nv_i2c_write_bits(0x08, ~(1<<2),(1<<2));//Open auto brake

haptic_nv_i2c_write(0x07, 0x80); //Set gain 0x80

haptic_nv_i2c_write(0x09, 0x01); //Start play
```

2.6.2 振动停止

1. 停止播放参考代码:

```
uint8_t reg_val = 0;



haptic_nv_i2c_write(0x09, 0x02);

for (int i = 0; i < 100; i++) {

    haptic_nv_i2c_read(0x3F, &reg_val); // Read global state register

    if (reg_val == 0x00) // Equal to 0 means that the chip has entered standby

        break;

    HAL_Delay(2); // Delay 2ms }

if (reg_val != 0) {

    haptic_nv_i2c_write_bits(0x44, ~(1<<6), (1<<6)); //force to standby

    haptic_nv_i2c_write_bits(0x44, ~(1<<6), (0<<6));

}
```

2.7 F0 校准

2.7.1 F0检测

1. F0 是马达共振时的频率。当驱动频率为F0 时，马达产生的振动量最大，因此线性马达要求驱动频率与马达的固有频率一致;

2．参考代码：

```
uint8_t reg_val = 0;
uint8_t brk_gain = 0x08;
uint8_t track_margain = 0x0F;
uint8_t drv2_lvl = 0;
uint32_t f0 = 0;
uint32_t f0_pre = 1700;
uint32_t vrms = 1000;
int drv_width = 0;

haptic_nv_i2c_write(0x5A, 0x00);// Set TRIM_LRA 0
haptic_nv_i2c_write_bits(0x08, ~(3<<0), (2<<0));      //Enter CONT mode
haptic_nv_i2c_write_bits(0x18, ~(1<<3), (1<<3));      //Enable F0_DET
haptic_nv_i2c_write_bits(0x1D, ~(1<<7), (1<<7));      //Enable track_en
haptic_nv_i2c_write_bits(0x08, ~(1<<2), (1<<2));      //Enable auto brake
haptic_nv_i2c_write_bits(0x1D, ~(0x7F<<0), 0x7F);    //Config cont_drv1_lvl
drv2_lvl = (((f0_pre < 1800) ? 1809920 : 1990912) / vrms * 61000000); // Calculate cont_drv2_lvl
```

```
if (drv2_lvl > 0x7F)
    drv2_lvl = 0x7F;
haptic_nv_i2c_write(0x1E, (uint8_t)drv2_lvl);          //Config cont_drv2_lvl
haptic_nv_i2c_write(0x1F, 0x04);                       //Config cont_drv1_time
haptic_nv_i2c_write(0x20, 0x06);                       //Config cont_drv2_time
haptic_nv_i2c_write (0x22, 0x0F);                      //Config cont_track_margin
drv_width = 240000 / f0_pre - 8 - brk_gain - track_margain; // Calculate cont_drv_width
if (drv_width < 0)
    drv_width = 0;
if (drv_width > 0xFF)
    drv_width = 0xFF;
haptic_nv_i2c_write(0x1A, drv_width);                  //Config cont_drv_width
haptic_nv_i2c_write(0x09, 0x01);                       //Start play
HAL_Delay(300);                                            //Delay 300ms;
haptic_nv_i2c_read(0x25, &reg_val);                    //LRA_F0 high 8 bits
f0 = reg_val << 8;
haptic_nv_i2c_read(0x26, &reg_val);                    //LRA_F0 low 8 bits
f0 = f0 | reg_val;
f0 = 384000 * 10 / f0;                                 // Calculate F0
printf("f0 is %d\r\n", f0);
haptic_nv_i2c_write_bits(0x18, ~(1<<3), (0<<3));       //Close F0_DET
haptic_nv_i2c_write_bits(0x08, ~(1<<2), (0<<2));       //Close auto brake
```

2.7.2 F0校准

1. 由于马达F0 的偏差，需要在生产线上整机情况下校准F0，保存校准值，并在播放时将其写入芯片;

2. 首先执行 F0 检测，获得马达的实际 f0，然后对其进行校准。参考校准代码如下所示:

```
char f0_cali_lra = 0;
int f0_cali_step = 0;
uint32_t f0_pre = 1700;
uint32_t f0_cali_min = 0;
uint32_t f0_cali_max = 0;

f0_cali_min = f0_pre * 93 / 100;
f0_cali_max = f0_pre * 107 / 100;
if (f0 < f0_cali_min || f0 > f0_cali_max)
    return;
f0_cali_step = 100000 * ((int)f0 - (int)f0_pre) / ((int)f0_pre * 24);
if (f0_cali_step >= 0) { /*f0_cali_step >= 0 */
if (f0_cali_step % 10 >= 5)
    f0_cali_step = 32 + (f0_cali_step / 10 + 1);
else
    f0_cali_step = 32 + f0_cali_step / 10;
}  else  { /* f0_cali_step < 0 */
if (f0_cali_step % 10 <= -5)
    f0_cali_step = 32 + (f0_cali_step / 10 - 1);
```

```
else
    f0_cali_step = 32 + f0_cali_step / 10;
}
if (f0_cali_step > 31)
    f0_cali_lra = (char)f0_cali_step - 32;
else
    f0_cali_lra = (char)f0_cali_step + 32;
printf("f0 cali data is 0x%02x\r\n", f0_cali_lra);
haptic_nv_i2c_write(0x5A, f0_cali_lra & 0x3F); //Write calibration value
```

3 参考函数

```
//i2c read byte
int haptic_nv_i2c_read(uint8_t reg_addr, uint8_t *reg_data)
{
   if (HAL_I2C_Mem_Read(&hi2c1, (0x58 << 1), reg_addr, I2C_MEMADD_SIZE_8BIT, reg_data, 1,
      1000) == HAL_OK)
            return 0;
      return -1;
}
//i2c read bytes
int haptic_nv_i2c_reads(uint8_t reg_addr, uint8_t *reg_data, uint16_t len)
{
if (HAL_I2C_Mem_Read(&hi2c1, (0x58 << 1), reg_addr, I2C_MEMADD_SIZE_8BIT, reg_data,
  len,1000) == HAL_OK)
            return 0;
      return -1;
}
//i2c write byte
int haptic_nv_i2c_write(uint8_t reg_addr, uint8_t reg_data)
{
if (HAL_I2C_Mem_Write(&hi2c1, (0x58 << 1), reg_addr, I2C_MEMADD_SIZE_8BIT, &reg_data, 1,
   1000) == HAL_OK)
```

```
        return 0;
    return -1;
}
//i2c write bytes
int haptic_nv_i2c_writes(uint8_t reg_addr, uint8_t *reg_data, uint16_t len)
{
    if (HAL_I2C_Mem_Write(&hi2c1, (0x58 << 1), reg_addr, I2C_MEMADD_SIZE_8BIT, reg_data,
        len, 1000) == HAL_OK)
        return 0;
    return -1;
}
//i2c write bits
void haptic_nv_i2c_write_bits(uint8_t reg_addr, uint32_t mask, uint8_t reg_data)
{
   uint8_t reg_val = 0;

   haptic_nv_i2c_reads(reg_addr, &reg_val, 1);
   reg_val &= (uint8_t)mask;
   reg_val |= (reg_data & (~mask));
   haptic_nv_i2c_writes(reg_addr, &reg_val, 1);
}
```

版本记录

版本	日期	说明
V1.0	2023-02-27	初版
V1.1	2023-09-05	修改芯片型号的区分说明

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